Central Geometry, Z Readouts, and Chiral Response in the Tri-Octagon Model

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Abstract

We consolidate a source-led investigation of the central twisted-hex research surface, the existing Z readouts, and the native three-channel update. The refined surface has an actual twelve-edge section whose area has a positive minimum; the surrounding host space has the complementary area maximum. Threefold fan openings and finite shell notches are distinct regions. Reversing an edge traversal, reflecting a mesh, and turning its zero member have different mathematical meanings. Under a declared spatial placement, the macro cone crosses each half of the finite band once per azimuth, but frozen-curve contact and discrete clock contact have different conditions. The full readout adds a chiral vector derived from the same native state. An inherited, subsequently audited four-update certificate illustrates an interior visit followed by three exterior observations, including exterior status at the first harmonic return. Exact symmetry residuals identify conditional correspondences and their obstructions. Finally, a norm budget and a weighted finite-step phase criterion explain distinct growth balances and opposite synchronization responses. Explicit finite phase domains give squared-chiral-magnitude gain and loss. Their proofs are separated from the inherited numerical enclosure chain locating the worked inputs within them; the scoped interval audit and exact replay are complete. No motion, flow, or geometry-to-dynamics coupling equation is assumed.

Preparation and status. Author and design testimony: Hilmir Frímann Halldórsson. Manuscript consolidation and written proof review: Codex, for GPT and Hilmir's review. This follows the attribution practice of the existing Z manuscript [4]; it does not imply approval of this draft. The ten controlling documents are unpublished investigation reports [15, 16, 17, 18, 19, 20, 21, 22, 23, 24]. The source snapshot actually read is `3b4aa9cb64b7c218ca82962dbf3caea39d9f7d34`.

Verification status. The numerical bounds originate in R3–R8. A subsequent standalone audit checked the required residual, propagation, finite-region, mechanism, and whole-domain phase inequalities using outward interval arithmetic and exact supporting identities [25]. GPT's isolated replay of the unchanged checker reproduced all 768 scalar records exactly; the scoped numerical review is closed [26]. Appendix F states its coverage and implementation assumptions. This v0.3 pass updates documents only. Author-wording approval and publication clearance remain separate.

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1 Questions, scope, and notation

The central object is an open, triangulated surface between two hexagonal rings, with three two-triangle fans above and three below. Its middle narrows, but the spaces on its two sides narrow in opposite senses: an inner section loses area while the surrounding section gains it. The Z readout is a separate construction from a complex three-channel state. Comparing its values with this object requires an explicit placement and a finite-region membership test. Neither a suggestive silhouette nor a sign change in a harmonic supplies that test.

Hilmir's narrowing, guidance, and forward/reverse descriptions motivate these questions. They are exploratory geometric pictures, not experimental evidence for fluid transport or an instruction to make the crystal move. Here "mirror" names a mathematical comparison operation; it does not prescribe a mirror-switching process. The paper asks what the existing definitions establish, including the cases in which a proposed correspondence fails.

We use the accepted width-one host [3, 6], the refined research family [7, 9], and the staged Z observer [4, 14]. The historical intersecting mesh is retained as history, not silently promoted to an embedded boundary. The native update is the existing polynomial-plus-synchronization map [5]. No part of the geometry enters that update. The spatial comparison below is conditional on the stated placement; physical flow would require additional equations and boundary data absent here. These qualifications apply throughout.

Symbol	Meaning
τ, ξ, δ	Crystal magnitude, $\xi = \tau/r$, and $\delta = \arctan \xi$; not clock time or a native coefficient.
w, η	Centered geometric height and section fraction $\eta = (1 - w/h)/2$.
$\mathcal{A}, \mathcal{A}_{\text{ext}}$	Inner and complementary section areas.
$\Omega = x + \mathrm{i}y,\ \kappa,\ \mathbf{C}$	Native state, its norm, and $x \times y$ in channel order $(23, 31, 12)$.
$\mathcal{P}, \epsilon, g, \lambda, k_j$	Native configuration and coefficients. λ is the synchronizer coefficient.
t_c, m, θ	Observer time, clock index, and clock angle.
K, A, b	Saturated macro amplitude before envelope, after envelope, and after an optional spatial scale.
$\mathbf{z}, \mathbf{M}$	Full Z observation and macro vector.
u, v, ζ	Two relative phases and their mean $\zeta = u + v/2$.
e_n, Q_n, D_n	Nominal proof error, original-box state error, and pre-synchronization box error; none changes the input radius.
H, S, Q, R	Height reflection, vertical reflection, their half-turn product, and axial 120° rotation.

Table 1: Notation used consistently in the manuscript. A source crosswalk is given in Appendix F.

Part I
What the central geometry provides

2 Construction, finite faces, and the two hands

2.1 What is inherited and what was chosen

The historical construction began with a gap triangle of folded octagons, translated it, made a second triangle, and made three rotational copies. The preserved rendering source subsequently added rings, nearest-apex cap triangles, and twisted side patches [8]. Its 18 vertices, 60 edges,

and 30 triangles form an open intersecting complex, not a closed polyhedron. Nearest-apex ties and the retained diameter faces matter to that distinction. The refined registration family omits the diameter faces, specifies symmetric fans, and replaces the historical side connectivity with the shifted-index band triangles defined in (2.3) [7]. Those choices are additional research definitions, not unique consequences of the outer octagons.

There is nevertheless an exact historical scale correspondence. With $b_h = \sin(\pi/8)$ and $c_h = \cos(\pi/8)$, the reconstructed primitive gap triangle has side lengths $(2b_h, 2b_h, \sqrt{2}b_h)$. Dividing by the historical octagon width $2c_h$ gives (s, s, d), where

$$s = \sqrt{2} - 1, \quad d = 1 - \frac{\sqrt{2}}{2}, \quad r = \frac{\sqrt{3}}{6}, \quad a = \frac{d}{2}, \quad h = \frac{s}{2}, \quad H_0 = \frac{1}{2}. \tag{2.1}$$

These are precisely the accepted host notch-triangle side lengths. Equality of these local triangles does not establish a rigid registration of the entire historical mesh [15, Parts A and C].

2.2 A complete definition of the refined surface

All coordinates below are centered at the host point $O = (0, r, 0)$. Define

$$\begin{gathered} \alpha_i = -\pi/2 + i\pi/3, \quad e_i = (\cos\alpha_i, \sin\alpha_i), \quad n_k = e_{2k}, \\ J(X, Y) = (-Y, X), \quad 0 \le \tau \le h, \quad \xi = \tau/r, \quad \delta = \arctan\xi, \\ A_\tau = I - \xi J = \sqrt{1+\xi^2}\, R_{-\delta}, \\ T_i = (ae_i, h), \quad B_i = (aA_\tau e_i, -h), \quad U_k = (rn_k, H_0), \quad L_k = (rA_\tau n_k, -H_0). \end{gathered} \tag{2.2}$$

Ring indices are modulo six and fan indices modulo three. The PLUS band and its upper fans consist of the finite triangles

$$\begin{aligned} F_i &= [T_i, T_{i+1}, B_{i+2}], & D_i &= [T_i, B_{i+2}, B_{i+1}], \\ \mho_{k,-} &= [U_k, T_{2k-1}, T_{2k}], & \mho_{k,+} &= [U_k, T_{2k}, T_{2k+1}]. \end{aligned} \tag{2.3}$$

The lower fans replace (U, T) by (L, B). Brackets denote the closed convex triangle, including its edges. The MINUS mesh is the image of this entire mesh under $S = \mathrm{diag}(-1, 1, 1)$, including its face list. It is not obtained by changing the sign of τ while retaining the PLUS faces.

This gives 18 vertices, 42 edges, and 24 faces, with an open annular topology and two boundary cycles [9]. The original candidate record used $0 < \tau \le h$; the later signed-family analysis explicitly adds $\tau = 0$. At zero the two hands have the same vertex set but distinct triangulated surfaces. The vertices have extra reflection symmetry which the actual faces do not inherit. The nonzero mesh has proper C_3 symmetry; the zero mesh has proper D_3 symmetry. Both remain chiral as triangulated surfaces [10, 11].

Proposition 2.1 (Axial clearance) *Every point of either refined mesh has cylindrical radius at least $a/2 > 0$. Every fan point satisfies $R_{\mathrm{cyl}} < |w|$.*

Proof. The planar vertices of each face lie in an angular wedge of width at most $2\pi/3$ and have radius at least a. Dotting with the wedge bisector gives at least $a\cos(\pi/3) = a/2$ for every convex combination. For the upper fans, the endpoint ratios are $a/h = 1/\sqrt{2}$ and $r/H_0 = 1/\sqrt{3}$. Convexity of the planar norm preserves the strict inequality relative to the linearly interpolated positive height. On lower fans the ratios are multiplied by $\sqrt{1+\xi^2}$. Their squared maxima satisfy

$$\frac{a^2(1+\xi^2)}{h^2} \le 5 - 3\sqrt{2} < 1, \qquad \frac{r^2(1+\xi^2)}{H_0^2} \le \frac{1}{3} + s^2 < 1.$$

Reflection does not change either radius or absolute height. □

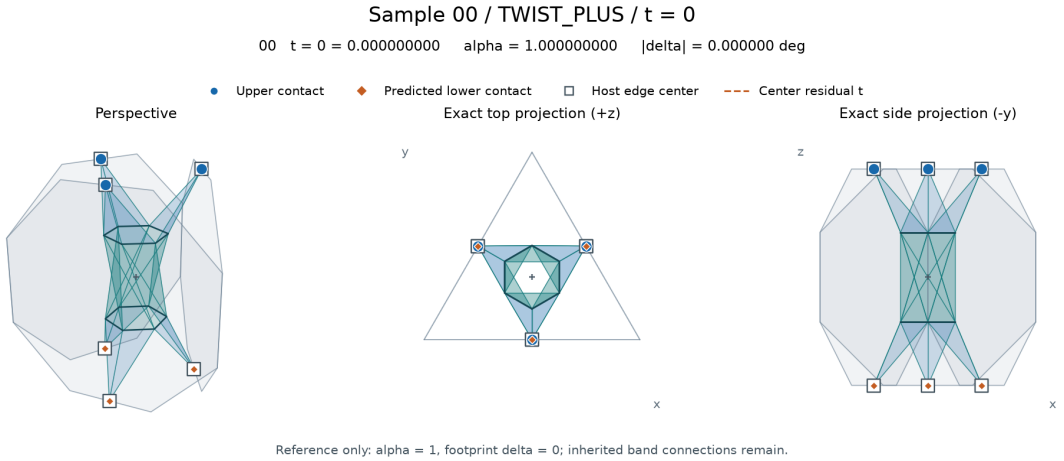


Figure 1: Retained static atlas image of the PLUS zero member, at the accepted width-one registration [12]. The atlas's t, α, δ mean this paper's crystal magnitude τ, inverse planar scale, and footprint angle. Thus $t = 0$ here, not observer time. The cyan finite surface retains the declared crossed band connections even though the ring footprints align. The grey host and contact markers supply registration context. Projection crossings are not a substitute for the face definitions in (2.3). No trajectory or mechanical turning path is depicted.

3 Actual sections and the finite constriction

Put $\eta = (1 - w/h)/2$. Intersecting the actual faces, rather than interpolating equal ring indices, gives

$$S_i(\eta) = (1-\eta)ae_i + \eta a A_\tau e_{i+1}, \qquad D_i(\eta) = (1-\eta)ae_i + \eta a A_\tau e_{i+2}. \tag{3.1}$$

At $0 < \eta < 1$, the twelve section edges are S_iD_i and D_iS_{i+1}. The section has C_6 symmetry although the complete surface has only C_3 in general. At the end levels these edges collapse in pairs to ring edges. The points $(1-\eta)ae_i + \eta a A_\tau e_i$ are not this section.

Theorem 3.1 (Section area and throat) *Every interior section is a simple star-shaped polygon about the axis. With $\mathcal{A}_0 = 3\sqrt{3}a^2/2$, its area is*

$$\mathcal{A}(\eta) = \mathcal{A}_0 \left[(1-\eta)^2 + (1+\xi^2)\eta^2 + \frac{4\xi}{\sqrt{3}}\eta(1-\eta) \right]. \tag{3.2}$$

Let $p = 1 - 2\xi/\sqrt{3}$ and $D = 2 + \xi^2 - 4\xi/\sqrt{3} = 2p + \xi^2$. Throughout $0 \le \tau \le h$, $p, D > 0$, and the unique minimum is

$$\eta_* = \frac{p}{D}, \qquad w_* = \frac{h\xi^2}{D}, \qquad \mathcal{A}_{\min} = \mathcal{A}_0 \left(1 - \frac{p^2}{D}\right) \ge \frac{\mathcal{A}_0}{2} > 0. \tag{3.3}$$

The formula and throat height are the same for both hands.

Proof. Write $b = a\sqrt{1+\xi^2}$. Consecutive determinants are

$$\det(S_i, D_i) = \frac{\sqrt{3}}{2}(\eta b)^2 + \eta(1-\eta)ab\sin\delta,$$

$$\det(D_i, S_{i+1}) = \frac{\sqrt{3}}{2}((1-\eta)a)^2 + \eta(1-\eta)ab\sin\delta.$$

They are positive for $0 < \eta < 1$, including $\xi = 0$. Their angular sectors assemble in the stated cyclic order with one winding, so the polygon is simple and each ray meets its boundary once. The

shoelace formula is three times their sum; $b \sin\delta = a\xi$ gives (3.2). The bracket is $1 - 2p\eta + D\eta^2$. Since $\xi \le \sqrt{3}s$, $p \ge 1 - 2s > 0$; also $D = 2p + \xi^2 > p$. Completing the square proves the unique interior minimum. Finally, $D - 2p^2 = 2p(1-p) + \xi^2 \ge 0$ because $0 < p \le 1$. □

At zero, $\eta_* = 1/2$ and $w_* = 0$. At $\tau = h$, $\eta_* = 1/5$, $w_* = 3h/5$, and $\mathcal{A}_{\min} = \mathcal{A}_0(1 - s^2/5)$. Thus a universal throat at the central plane would be incorrect.

3.1 Inner and surrounding spaces

In the slab $|w| < h$ the host cross-section is an equilateral support triangle with inradius r and area $\sqrt{3}/4$. The band section lies strictly inside its incircle because $a\sqrt{1+\xi^2} < h < r$. Consequently

$$\mathcal{A}_{\text{ext}}(w) = \frac{\sqrt{3}}{4} - \mathcal{A}(w) \tag{3.4}$$

has a maximum at the inner throat. This surrounding cross-section is a connected annular region, not three disconnected channels.

An area constriction is therefore established for the inner finite slab. A Venturi or Bernoulli conclusion is not: there is no specified fluid, pressure, conserved flux, impermeable wall model, or inlet/outlet problem. The surface is open, and its fan openings are additional geometry outside the central slab. Area alone determines neither velocity nor transport direction. This is the precise geometric content of the passage analogy.

4 Fans, openings, and finite shell notches

Fan material, an opening between neighboring fans, a shell notch measurement patch, and a material shell panel are four different sets. Put $g_k = e_{2k+1} = n_k + n_{k+1}$ and $j_k = Jg_k$. The finite shell notch patch is

$$\begin{aligned} N_{k,\sigma}(\lambda_N, \zeta_N) = \left(2r - \frac{\sqrt{3}d}{2}\lambda_N\right) g_k + \zeta_N j_k + \sigma(h + d\lambda_N)e_z, \\ 0 \le \lambda_N \le 1, \qquad |\zeta_N| \le d\lambda_N/2, \qquad \sigma = \pm 1. \end{aligned} \tag{4.1}$$

Its plane is

$$g_k \cdot (X, Y) + \sigma\frac{\sqrt{3}}{2}w = 2r + \frac{\sqrt{3}}{2}h. \tag{4.2}$$

The displayed plus sign before $\zeta_N j_k$ is essential. A point on this infinite plane is not necessarily on the finite notch.

The upper opening boundary is the broken arc $U_k - T_{2k+1} - U_{k+1}$. Its two side points at height $h + d\lambda_N$ are

$$[(1-\lambda_N)a + \lambda_N r/2]\, g_k \ \pm \lambda_N\sqrt{3}r\, j_k/2.$$

Its azimuthal bisector is g_k, the same as the upper notch, whereas the fan tip is on n_k. These are alternating sets of three directions. The lower opening is rotated by $-\delta$ and dilated by $\sqrt{1+\xi^2}$ in PLUS; the host notch is fixed. Alignment of upper azimuths therefore does not imply coincidence of the finite regions, and lower alignment varies with τ.

For incidence, substitute vertices into the linear left side of (4.2). Upper fans are strictly on the inner side of every upper notch plane. The maximum lower-tip value leaves the nonnegative gap $\sqrt{3}(h - \tau)/2$; all lower-ring vertices are strictly inside. Thus lower fans do not meet lower notch patches for $\tau < h$. At $\tau = h$ the relevant lower tips meet notch vertices; there is no positive-area fan/notch coincidence. Opposite-level patches are excluded by height. This vertex-and-convexity argument concerns finite patches as defined in (4.1); it does not turn a measurement triangle into material [17, R2, Sections 3–4].

5 Traversal reversal and actual geometric operations

5.1 Reversing a line does not change its hand

Each band cross-edge is

$$\Gamma_{i,m}(\eta) = ((1-\eta)ae_i + \eta aA_\tau e_{i+m},\ h(1-2\eta)), \qquad m = 1, 2.$$

Its planar component never vanishes. On a continuous azimuth lift,

$$\frac{d\phi}{d\eta} = \frac{a^2\sqrt{1+\xi^2}\sin(m\pi/3-\delta)}{|(1-\eta)ae_i + \eta aA_\tau e_{i+m}|^2} > 0, \qquad \frac{dw}{d\eta} = -2h. \tag{5.1}$$

Here $0 \le \delta < \pi/3$. Traversing the same edge with decreasing η reverses both derivatives and preserves their ratio. No reflection or change of τ is needed. The same distinction applies to reversing a piecewise fan-boundary arc.

5.2 Face-level actions at zero and their obstruction away from zero

Use the centered matrices $H = \mathrm{diag}(1, 1, -1)$, $S = \mathrm{diag}(-1, 1, 1)$, $Q = SH = \mathrm{diag}(-1, 1, -1)$, and $R = R_{2\pi/3}$ about the axis. At $\tau = 0$, Q is a proper half-turn preserving each hand, while H and S exchange hands. Actual face correspondences, with a superscript minus denoting the reflected mesh, are

	F_i	D_i
H	D^-_{i+2}	F^-_{i+2}
S	F^-_{-i}	D^-_{-i}
Q	D_{-i-2}	F_{-i-2}.

(5.2)

These follow by applying each matrix to the three vertices in (2.3), using $Se_i = e_{-i}$, and comparing the unordered triples. For example $Q\Gamma_{i,m}(\eta) = \Gamma_{-i-m,m}(1-\eta)$. Q sends upper fan k to lower fan $-k$ and upper opening arc k to the reversed lower arc $-k-1$. On finite notches,

$$H : (k, \sigma) \mapsto (k, -\sigma), \quad S : (k, \sigma) \mapsto (-k-1, \sigma), \quad Q : (k, \sigma) \mapsto (-k-1, -\sigma), \quad R : k \mapsto k+1.$$

Substitution into (4.1) preserves both parameter inequalities.

For $\tau > 0$, either H or Q would put the lower ring of radius $a\sqrt{1+\xi^2} > a$ at the upper height. Every member in the registered family has upper radius a. Hence that image is not another registered member. The mirror S still exchanges hands at the same τ.

Finally, an axial rotation and translation preserving the exact upper three contacts must have translation zero and rotation angle in $2\pi\mathbb{Z}/3$. Continuity leaves only a constant such registered motion. This is a constraint on a proposed motion with all contacts retained, not a statement that the contacts are welded or a dynamical law. The zero half-turn is an endpoint symmetry; it does not supply a feasible continuously registered turning path [24, R9, Sections 2–3].

Part II
How Z observations compare with the geometry

6 One native state, the staged observer, and placement

6.1 Native update and readout definitions

Let $\Omega = (\Omega_1, \Omega_2, \Omega_3) = x + \mathrm{i}y \in \mathbb{C}^3$ and $\mathcal{D} = (\epsilon, g, \lambda, k_1, k_2, k_3)$. With $\mathbf{L}$ having diagonal -2 and off-diagonal 1, the complete update is

$$\begin{aligned} V = \mathcal{G}_{\mathcal{D}}(\Omega) &= \Omega + \epsilon\,\Omega \odot (k - |\Omega|^2) + g\mathbf{L}\Omega, \\ \Delta_j &= \lambda \sum_{m \neq j} \sin(3(\phi_m - \phi_j)), \qquad \phi_j = \operatorname{Arg} V_j, \\ \Omega_j^+ = \mathcal{F}_{\mathcal{D}}(\Omega)_j &= |V_j| \exp(\mathrm{i}(\phi_j + \Delta_j)). \end{aligned} \tag{6.1}$$

All Δ_j use the same pre-synchronization phases. The source convention is $\operatorname{Arg} 0 = 0$, including that phase in neighboring sums. When $\lambda = 0$ the source bypasses synchronization. The nonzero-component arguments later in the paper explicitly state their domains. Observer time increments are not factors in (6.1) [1, 5].

Define

$$\begin{gathered} \kappa = \|\Omega\|, \qquad \mathbf{C} = x \times y = (\Im(\overline{\Omega}_2\Omega_3), \Im(\overline{\Omega}_3\Omega_1), \Im(\overline{\Omega}_1\Omega_2)), \\ K(\kappa) = 0.618\,\frac{\kappa}{1+\kappa}, \quad A(\kappa, t_c) = K(\kappa)\mathrm{e}^{-0.577 t_c}, \quad f(\theta) = \cos 3(\theta - \ell), \quad \ell = 0.244, \\ d_\theta = (\cos\theta, \sin\theta, 1), \qquad \mathbf{M} = A f(\theta) d_\theta, \qquad \mathbf{z} = \alpha\mathbf{M} + \beta\mathbf{C}. \end{gathered} \tag{6.2}$$

The existing weights are $\alpha = 1, \beta = 1/2$. The macro amplitude and the chiral vector depend on the *same* state; they are not freely adjustable inputs to a fit. The area identity

$$\|\mathbf{C}\|^2 = \|x\|^2 \|y\|^2 - (x \cdot y)^2 \le \kappa^4/4$$

is the Gram determinant and the elementary arithmetic–geometric mean inequality [2, 13]. The dimensionless passive ratio $\chi = 2\|\mathbf{C}\|/\kappa^2$ is defined for $\kappa > 0$.

The staged schedule uses twelve sectors, $m_n = n \bmod 12$, $\theta_n = m_n\pi/6$, and $t_{c,n} = 0.1n$. First observe the given state at $n = 0$; then update the state, advance the clock and time, and observe. Thus the nth scheduled observation uses exactly n native updates. The committed historical EMA observer is a different law and is not used here [4, 14].

6.2 Spatial meaning is declared

The comparisons in Sections 7–9 use

$$\Phi(\mathbf{z}) = O + \mu R_\psi \mathbf{z}, \qquad \mu > 0, \tag{6.3}$$

with an axial rotation R_ψ. Centered height is the third component of $\mu R_\psi \mathbf{z}$. The worked certificate fixes $\mu = 1, \psi = 0$ and PLUS $\tau = 0$. Channel indices in Ω are not automatically spatial axes under every shell symmetry; Section 10 tests that issue. Anisotropic fitting is not introduced.

7 Cone contact, frozen curves, and discrete samples

Theorem 7.1 (Cone–band crossing) *For either hand and every $\tau \in [0, h]$, the double cone $R_{\mathrm{cyl}} = |w|$ misses all fan triangles and meets the band once on each half-ray $(t\cos\varphi, t\sin\varphi, \sigma t)$, $t > 0$, $\sigma = \pm 1$. Write this unique radius as $T_\sigma(\varphi) \in (0, h)$. It is continuous and C_6-periodic.*

Proof. Fan exclusion is Proposition 2.1. Write $\mathcal{R}(w, \varphi) > 0$ for the radial boundary of the star-shaped section. At $w = 0$ it is positive; at $w = \sigma h$ it is less than h, since the largest ring radius is less than h. Continuity therefore gives a crossing. For uniqueness, a section edge lies in a band face plane $n_{xy} \cdot (X, Y) + n_w w = c_F$. With $k_0 = \sqrt{3}/2$, valid outward data for PLUS face index i are

	n_{xy}	n_w	c_F
F_i	$R_{\alpha_i}(k_0, 1/2)$	$a(\xi - k_0)/(2h)$	$a(k_0 + \xi)/2$
D_i	$R_{\alpha_i}(\xi, 1)$	$a[k_0(1+\xi^2) - \xi]/(2h)$	$a[k_0(1+\xi^2) + \xi]/2.$

(7.1)

The constants and the supporting radial denominators $n_{xy} \cdot e_\varphi$ are positive on their section edges. On each such piece,

$$\frac{\mathcal{R}(\sigma t, \varphi)}{t} = \frac{c_F}{(n_{xy} \cdot e_\varphi)t} - \frac{\sigma n_w}{n_{xy} \cdot e_\varphi}$$

is strictly decreasing in t. It is continuous across edge changes; the finite pieces establish uniqueness. Compactness and this unique crossing give continuity; the sixfold section symmetry gives periodicity. Reflection proves the other hand. □

This also corrects any argument based on corresponding-index ring interpolation: the existence conclusion survives, but its proof uses the actual triangles. Contact is with a finite face. For a candidate cone point in a particular face,

$$t = \frac{c_F}{n_{xy} \cdot e_\varphi + \sigma n_w} > 0$$

must additionally satisfy that triangle's barycentric inequalities (Appendix A).

Theorem 7.2 (Frozen and sampled contact are different) *Freeze the native state and observer time with nonzero macro amplitude, put $b = \mu\alpha A > 0$, and $\vartheta = \ell + \psi$. The placed signed macro curve meets band half σ exactly when*

$$b \geq B_\sigma, \qquad B_\sigma = \min_{\cos 3(\varphi - \vartheta) > 0} \frac{T_\sigma(\varphi)}{\cos 3(\varphi - \vartheta)}. \tag{7.2}$$

The minimum is attained and $a/2 < B_\sigma < h$. On the twelve-sector schedule with the default lock, the set of amplitudes giving at least one exact sampled contact is the finite set

$$\bigcup_{\sigma = \pm 1} \left\{ \frac{T_\sigma(\psi)}{\cos(0.732)}, \frac{T_\sigma(\psi + \pi/6)}{\sin(0.732)} \right\}. \tag{7.3}$$

These conditions do not assert that every mathematical amplitude is reachable under the bounded source amplitude and a fixed placement scale.

Proof. For either cone half, use the actual spatial azimuth. When the signed scalar is negative this is $\varphi = \theta + \psi + \pi$, not $\theta + \psi$. In both halves $t = b\cos 3(\varphi - \vartheta) > 0$. On each positive-harmonic interval the quotient in (7.2) is continuous, tends to infinity at both ends, and has image $[B_\sigma, \infty)$. Threefold repetition gives the same minimum. Clearance gives the lower bound; equality at $a/2$ would require a limiting wedge midpoint at $w = 0$, which is not a nonzero cone contact. At a harmonic maximum the quotient is $T_\sigma < h$, proving the upper bound. Above the minimum there

are at least two roots per positive interval, and at the minimum at least one. On the clock, f repeats $(c, s_\ell, -c, -s_\ell)$, where $c = \cos(0.732) > 0$ and $s_\ell = \sin(0.732) > 0$. Sixfold periodicity of T_σ reduces all twelve samples to (7.3). □

For example, if $b > h/\min(c, s_\ell)$ every discrete sample is beyond the band although the continuous frozen curve has contacts. At $\tau = 0, \psi = 0$ a concrete finite-face contact is available. Put $k_* = \sqrt{6}/8$, $c_* = a\sqrt{3}/4$ (only in this paragraph), $t_* = c_*/(1 - k_*)$, and $\eta_* = (1 - t_*/h)/2 < 1/3$. Then $(t_*, 0, t_*) \in F_1$ with barycentrics $((1 + \eta_*)/2, (1 - 3\eta_*)/2, \eta_*)$; the negative-height point is in D_0 with the cyclic weights stated in Appendix A. Taking $b = t_*/c$ gives sampled contact. It is not the continuous threshold: the quotient derivative at azimuth zero is $-3t_* s_\ell/c^2 < 0$ [16, R1, Sections 4–6].

8 Full Z and the exact same-state contact witness

For a frozen state and time, full Z translates the macro curve by $\Delta_C = \mu\beta R_\psi \mathbf{C}$; its tangent with respect to angle is unchanged by this translation. Consequently it need not lie on the original macro cone, and the fan exclusion does not apply to it. A face contact solves

$$\mu\alpha A f(\theta)\, n \cdot R_\psi d_\theta = c_F - n \cdot \Delta_C \tag{8.1}$$

together with finite-face barycentrics. The undivided equation includes zero harmonics and zero denominators. All harmonic zeros map to Δ_C, not necessarily to the origin.

The following construction realizes a fan point with the actual coupled state variables; it does not vary A and $\mathbf{C}$ independently. At PLUS zero with identity placement, let

$$\begin{gathered} P = \left(\frac{\sqrt{3}a}{6}, -\frac{a}{2} - \frac{r}{3}, \frac{2h + H_0}{3}\right), \qquad A_* = 0.4635, \qquad c = \cos(0.732), \\ c^\star = 2\{P - A_* c(1,0,1)\}, \qquad L = \|c^\star\|, \qquad b_\perp = \sqrt{(c_1^\star)^2 + (c_2^\star)^2}, \\ p_0 = (-c_2^\star, c_1^\star, 0)/b_\perp, \qquad q_0 = (c^\star/L) \times p_0, \\ X = \sqrt{\frac{9 + \sqrt{81 - 4L^2}}{2}}, \qquad Y = \sqrt{\frac{9 - \sqrt{81 - 4L^2}}{2}}, \qquad \Omega_* = Xp_0 + \mathrm{i}Yq_0. \end{gathered} \tag{8.2}$$

Here $0 < L < 4 < 9/2$ and $b_\perp > 0$; the simple bounds in Appendix B also certify them. The vectors p_0, q_0 are orthonormal, $X^2 + Y^2 = 9$, and $XY = L$, so

$$\|\Omega_*\| = 3, \qquad \mathbf{C}(\Omega_*) = c^\star, \qquad \mathbf{z}(\Omega_*, 0, 0) = A_* c(1,0,1) + c^\star/2 = P.$$

P is the centroid of $[U_0, T_0, T_1]$, with all three barycentric weights $1/3$. This is an engineered exact witness, not a reconstructed historical trajectory [17, R2, Section 7].

There is no universal extension of the macro amplitude threshold to full Z: under scaling of a fixed state, the saturated macro factor stays bounded while $\mathbf{C}$ scales quadratically. The same-state constraint remains essential even when it permits a contact.

9 The original input box through four updates

9.1 Hypotheses and the compact certificate

Use the exact state (8.2), not a decimal representative, and reference tuple

$$\mathcal{P}_* = (0.1, 0.2, 0.17, 2, 3, 4).$$

Let $\mathcal{B}$ be the closed Cartesian product in the six real state coordinates $(x_1, x_2, x_3, y_1, y_2, y_3)$ and the six configuration coordinates: every coordinate is within 10^{-8} of its reference value. Each chosen $\mathcal{P}$ is held identical across all updates. Geometry, placement, observer weights, lock, and schedule are fixed. In particular $\tau = 0$, so the throat plane is $w = 0$. Exact initial fan contact is required only of the center, not of every perturbed input.

Proposition 9.1 (Audited uniform four-update certificate) *For each input of $\mathcal{B}$, apply exactly four complete native updates (6.1). The resulting observations satisfy Table 2. Every observation after the initial one is inside the host's central triangular prism; only observation one is inside the band interior. Observations two, three, and four lie strictly outside its finite boundary. In particular, the positive harmonic at observation four does not restore band-interior membership.*

n	$f(\theta_n)$	Signed height w_n	Axis distance	Center distance	Band
0	c	$(.3046, .3049)$	$(.1693, .1760)$	$(.3484, .3530)$	above slab
1	s_ℓ	$(.16734, .16788)$	$(.10550, .1090)$	$(.1970, .2010)$	inside
2	$-c$	$(-.1102, -.1075)$	$(.155, .160)$	$(.1886, .195)$	outside
3	$-s_\ell$	$(-.0939, -.0926)$	$(.1387, .1401)$	$(.1667, .1687)$	outside
4	c	$(.1333, .1370)$	$(.1247, .1301)$	$(.1825, .189)$	outside

Table 2: Uniform strict rational enclosures for the unchanged closed input box. Throat-plane distance is $|w_n|$, not signed height. The first row permits nearby points off the fan; its central input is the exact contact witness. The first three rows inherit R4–R5, the last two R6; their scoped numerical checks are complete [25, 26].

Proof and completed numerical check. Appendix B gives the rational comparison states, residual allowances, Lipschitz domains, and recurrences from R3–R6. The standalone audit independently checked these inherited enclosures and their coverage on the original closed box [25]. The recurrences bound every input, with its configuration fixed through all four updates. Appendix A carries the resulting coordinates through finite section tests, including the directional test needed at observation four. Those strict inequalities prove the classifications, not the harmonic signs. The unchanged-checker replay reproduced the scalar certificates exactly [26]; the implementation qualifications in Appendix F still apply. □

All distance comparisons are with each input's own preceding observations. From two to three, distances to the throat, axis, and center all decrease. From three to four, the axis distance decreases but the throat and center distances increase. Relative to observation one, observation four is closer to the throat and center but farther from the axis. All four scheduled endpoints are closer to the throat than their own initial observations. These claims follow from disjoint intervals, without comparing different inputs at selected midpoint values.

The first update also separates a native-only endpoint from a scheduled endpoint. Let $\mathbf{z}_N = \mathbf{z}(\Omega^+, t_c = 0, \theta = 0)$. Uniformly it has $w_N \in (.19192, .19245)$ and radius greater than $a + .00197$, while the scheduled endpoint has radius less than .109 and lies inside a section of inradius greater than .1144. Both remain above and closer to $w = 0$ than their own initial observation: $w_0 > w_N > w_1 > 0$. This is the local robustness result of R4, with no exact initial contact imposed on perturbations.

The table is a local finite certificate. It is not a basin, a general stability result, or a continuous curve passing through successive openings. No interpolation between observations is a native trajectory in the source.

9.2 Native, envelope, and clock contributions

Let $E = e^{-.0577}$, $K_n = K(\kappa_n), f_n = f(\theta_n)$, and $d_n = d_{\theta_n}$. The exact telescoping split is

$$\begin{aligned}\mathbf{z}_{n+1} - \mathbf{z}_n = {} & E^n f_n (K_{n+1} - K_n) d_n + \frac{\mathbf{C}_{n+1} - \mathbf{C}_n}{2} \\ & + K_{n+1} E^n (E-1) f_n d_n \\ & + K_{n+1} E^{n+1} (f_{n+1} d_{n+1} - f_n d_n).\end{aligned} \tag{9.1}$$

These terms change the norm-derived amplitude, then chirality, then the envelope, then the clock. The first two together are the native-only change. Using the inherited bounds, their vertical sum is $(-.00549, -.00395)$ from two to three and $(-.03027, -.02646)$ from three to four. The corresponding envelope changes are $(.00747, .00754)$ and $(.00697, .00707)$; the clock changes are $(.01277, .01287)$ and $(.2485, .2509)$. In both transitions the native-only vertical change is negative. The scheduled increases in signed height are therefore not native upward forcing. In the fourth observation the positive clock term is decisive for the sign, while finite geometry still determines exterior status.

10 Symmetry residuals and conditional correspondences

10.1 Macro identities at equal state and time

Freeze Ω, t_c and write $A = A(\kappa, t_c)$. Direct trigonometry gives

$$\begin{aligned}\mathbf{M}(\theta + \pi) &= H\mathbf{M}(\theta), & \mathbf{M}(\theta + 2\pi/3) &= R\mathbf{M}(\theta), \\ \mathbf{M}(\pi - \theta) - S\mathbf{M}(\theta) &= -2Ac \cos 3\theta \, d_{\pi-\theta}, \\ \mathbf{M}(-\theta) - Q\mathbf{M}(\theta) &= 2Ac \cos 3\theta \, d_{-\theta}.\end{aligned} \tag{10.1}$$

On the twelve-sector grid these relabel m as $m + 6, m + 4, 6 - m, -m$. The latter two residuals vanish at odd sectors and generally not at even ones. Reflection through the lock-dependent angle ℓ instead gives an exact continuous macro identity, but its clock index is $12\ell/\pi - m$, not an integer with the default lock. Its axes also fail to be the shell reflection axes. Reversing the parameter of any frozen curve always reverses its traversal; it does not require a spatial symmetry.

10.2 Native channel actions are separate objects

Let $\mathcal{K}\Omega = \overline{\Omega}$, let P swap channel slots one and three, and let $P_c(\Omega_1, \Omega_2, \Omega_3) = (\Omega_3, \Omega_1, \Omega_2)$. The shell-induced actions from the controlling channel convention are

$$T_H = \mathcal{K}, \qquad T_S = -P\mathcal{K}, \qquad T_Q = -P, \qquad T_R = P_c.$$

Their chiral vectors are respectively $-\mathbf{C}$, $P\mathbf{C}$, $-P\mathbf{C}$, and $P_c\mathbf{C}$, whereas the geometric actions are $H\mathbf{C}, S\mathbf{C}, Q\mathbf{C}, R\mathbf{C}$. The distinction is an ordinary proper/improper transformation issue: for a real matrix B, $(Bx) \times (By) = \operatorname{cof}(B)(x \times y)$.

Conjugation commutes with the complete forward map at the same real configuration, including the zero convention. Simultaneously permuting the channels and their k labels is a covariance. At a fixed configuration, the mismatch below vanishes in particular when $Pk = k$ or $\epsilon = 0$; a general fixed-configuration permutation symmetry is not supplied by the worked unequal-k tuple:

$$\mathcal{G}_{\mathcal{P}}(P\Omega) - P\mathcal{G}_{\mathcal{P}}(\Omega) = \epsilon \, P\Omega \odot (k - Pk).$$

Vanishing on a particular state does not establish a general covariance. Multiplication by a common phase, in particular a minus sign, commutes with synchronization on nonzero pre-sync components (or with the bypass $\lambda = 0$); it is not a blanket zero-component statement with $\operatorname{Arg} 0 = 0$.

Cross-product transformation identities themselves do include zeros. Finally, $\mathcal{K}\mathcal{F}\mathcal{K} = \mathcal{F}$, not $\mathcal{F}^{-1}$: forward-map commutation is not inverse-time evolution.

Proposition 10.1 (Exact full-Z residuals) *At equal observer time, with the angle relabels of* (10.1)*, define* $\mathcal{E}_G = \mathbf{z}(T_G\Omega, \hat{\theta}) - G\mathbf{z}(\Omega, \theta)$*. Then*

$$\begin{aligned} \mathcal{E}_H &= (-2\beta C_1, -2\beta C_2, 0), \\ \mathcal{E}_Q &= 2\alpha Ac \cos 3\theta\, d_{-\theta} + \beta(C_1 - C_3, -2C_2, C_3 - C_1), \\ \mathcal{E}_S &= -2\alpha Ac \cos 3\theta\, d_{\pi-\theta} + \beta(C_1 + C_3, 0, C_1 - C_3), \\ \mathcal{E}_R &= \beta(P_c\mathbf{C} - R\mathbf{C}). \end{aligned} \tag{10.2}$$

Here (C_1, C_2, C_3) *are components of the single original state.*

Proof. All listed state actions preserve κ, so they preserve A. Subtract the geometric action from the new full observation, use (10.1), and insert the four chiral transformations above. This proves the residuals without assuming a new native orbit. □

For H with conjugation, an equal-time match holds exactly when $C_1 = C_2 = 0$; using the *same* state instead requires $C_3 = 0$. For Q with its shell action $-P$, odd grid sectors match exactly when $C_1 = C_3$ and $C_2 = 0$. For S, odd sectors require $C_1 = C_3 = 0$. For the rotational generator R, equality holds precisely on the line $\mathbf{C} = t(-(2 + \sqrt{3}), 1, 1)$; simultaneous matching of both R and R^2 requires $\mathbf{C} = 0$.

A different comparison, Q with conjugation, is useful but is not the shell-induced state action:

$$\mathbf{z}(\mathcal{K}\Omega, -\theta) - Q\mathbf{z}(\Omega, \theta) = 2\alpha Ac \cos 3\theta\, d_{-\theta} - (0, 2\beta C_2, 0). \tag{10.3}$$

For a nonzero state with the present positive weights, it vanishes on the grid exactly at odd sectors with $C_2 = 0$. This provides a conditional same-hand zero-mesh bridge. It does not supply a later observation: the clock relabel $m \mapsto -m$ reverses order and does not commute with the unchanged increment $+1$.

Appendix E records the exceptional even-sector cancellations. They are pointwise conditions, not all-grid symmetries. The original witness has $C_2 = -a - 2r/3 < 0$ and $C_3 < 0$, and it occurs at sector zero. It therefore does not satisfy the H bridges or the Q conjugation bridge; the shell-action Q match at sector zero is impossible for any nonzero state. It is not legitimate to replace that witness by a different state satisfying the desired match.

If a residual vanishes and an observation already lies on a specified finite face or notch, applying the geometric map transports its barycentric or notch parameters and proves membership in the mapped finite region. This is a contact implication. It neither establishes the existence of such a contact under the special chiral conditions nor selects an exit or a physical direction of travel.

Part III
Why the native contributions change

11 Norm and chiral-area mechanisms

11.1 The complete finite norm budget

Set $I_j = |\Omega_j|^2$, $H_\Omega = \Omega \odot (k - I)$, $D_\Omega = \epsilon H_\Omega + g\mathbf{L}\Omega$, and $J_\Omega = \sum_{j<k} |\Omega_j - \Omega_k|^2$. Synchronization preserves each pre-sync magnitude, so expansion of the squared norm gives the exact identity

$$\kappa_+^2 - \kappa^2 = \underbrace{2\epsilon \sum_j I_j(k_j - I_j)}_{\mathcal{O}} \underbrace{- 2gJ_\Omega}_{\mathcal{G}} + \underbrace{\|D_\Omega\|^2}_{\mathcal{R}}. \tag{11.1}$$

Here $\Re \langle \Omega, \mathbf{L}\Omega \rangle = -J_\Omega$. The nonnegative finite-increment term must be retained; this is a discrete identity, not a first-order differential approximation. Further,

$$\mathcal{R} = \epsilon^2 \|H_\Omega\|^2 + 2\epsilon g \Re \langle H_\Omega, \mathbf{L}\Omega \rangle + g^2 \|\mathbf{L}\Omega\|^2, \qquad \|\mathbf{L}\Omega\|^2 = 3J_\Omega. \tag{11.2}$$

The last identity follows from $\mathbf{L}^2 = -3\mathbf{L}$. These are standard norm expansions, already used for this model in [4, Section 17] and implemented as diagnostics in [5].

Transition	$\mathcal{O}$	$\mathcal{G}$	$\mathcal{R}$	$\kappa_{n+1}^2 - \kappa_n^2$
$1 \to 2$	$(.16900, .16924)$	$(-.27310, -.27283)$	$(.03635, .03653)$	$(-.06775, -.06706)$
$2 \to 3$	$(.14654, .14711)$	$(-.16500, -.16426)$	$(.02510, .02562)$	$(.00664, .00847)$
$3 \to 4$	$(.15774, .16009)$	$(-.10210, -.09942)$	$(.02250, .02509)$	$(.07814, .08576)$

Table 3: Uniform inherited R7 norm budgets on $\mathcal{B}$, with the complete fixed-per-input update. Allowances are given in Appendix C.

The first transition contracts because its pairwise penalty dominates. From two to three, $\mathcal{O} + \mathcal{G} \in (-.01846, -.01715)$ is still negative, and the squared finite increment is essential to growth. From three to four, $\mathcal{O} + \mathcal{G} \in (.05564, .06067)$ is already positive; the remainder increases it further. The dominant onsite contribution is channel three, followed by two and one. The two large pair differences involving channel one decrease strongly before the latter transition; the small 23 difference increases. The mixed term in (11.2) is negative in all three rows, so the remainder is not interpreted as a sum of three positive mechanisms.

11.2 Pre-sync transport and simultaneous phase change

At a fixed state, the polynomial stage is $V = \mathbf{B}\Omega$ with the real matrix

$$\mathbf{B} = \operatorname{diag}(1 + \epsilon(k - I)) + g\mathbf{L}.$$

It follows exactly that

$$\mathbf{C}(V) = \operatorname{cof}(\mathbf{B})\mathbf{C}(\Omega). \tag{11.3}$$

The cofactor identity is valid even when $\mathbf{B}$ is singular. $\mathbf{B}$ depends on the state and is not the Jacobian of the nonlinear update.

For a cyclic pair (j, k) write $\overline{V}_j V_k = P_{jk} + \mathrm{i}A_{jk}$ and $d_{jk} = \Delta_k - \Delta_j$. Simultaneous synchronization gives

$$C_i^+ = A_{jk} \cos d_{jk} + P_{jk} \sin d_{jk}, \qquad C_i^+ - C_i(V) = A_{jk}(\cos d_{jk} - 1) + P_{jk} \sin d_{jk}. \tag{11.4}$$

State	C_3	$\|\mathbf{C}\|$	$\chi = 2\|\mathbf{C}\|/\kappa^2$	$C_3/\|\mathbf{C}\|$
Ω_2	$(.04689, .04712)$	$(.0745, .0749)$	$(.652, .659)$	$(.627, .632)$
V_3	$(.03798, .03827)$	$(.0622, .0627)$	$(.527, .534)$	$(.607, .614)$
Ω_3	$(.04008, .04101)$	$(.0649, .0665)$	$(.550, .567)$	$(.608, .626)$
V_4	$(.03237, .03394)$	$(.0537, .0564)$	$(.335, .359)$	$(.584, .621)$
Ω_4	$(.00515, .00924)$	$(.0086, .0150)$	$(.053, .096)$	$(.592, .620)$

Table 4: Inherited R7 stage separation. C_3 is a signed component; $\|\mathbf{C}\|$ is full magnitude. The last two columns describe scale-free area and one directional projection, respectively.

This separates polynomial changes from the rotation of all pair products by the already computed phase increments.

In update three, the polynomial stage lowers C_3 by $(-.00914, -.00862)$ and synchronization raises it by $(.00218, .00268)$; the net change remains $(-.00696, -.00594)$. Full magnitude also decreases before synchronization and increases during it, with a net decrease. In update four, both stages lower C_3, by $(-.00864, -.00614)$ and $(-.02800, -.02400)$, giving $(-.03586, -.03084)$. Full magnitude decreases in both stages. These conclusions survive division by the positive stage norms as shown by the χ column.

Direction is a separate question. The separated ratios in Appendix C establish changes of direction at each of the two stages. The complete transition from two to three decreases the vertical fraction and therefore changes direction. The sign of the vertical-fraction change during the third synchronization, during either stage of the fourth update, and over the complete fourth update remains unresolved by the inherited bounds. The net direction change from three to four is also unresolved: two stage changes could partly undo each other. Overlapping intervals do not show mixed behavior. In particular, a falling C_3 cannot by itself be described as either a pure rotation or a pure loss of area. Equations (9.1) and (11.4) keep these native effects separate from the observer clock.

12 Finite phase domains for synchronization gain and loss

12.1 Exact two-phase reduction

Assume $V_j = r_j e^{i\phi_j}$ with every $r_j > 0$, and choose continuous local phase lifts. Define $u = \phi_2 - \phi_1$, $v = \phi_3 - \phi_2$, and $\zeta = u + v/2$. Subtracting the simultaneous increments in (6.1) gives

$$\begin{aligned} u' &= u + \lambda[-2\sin 3u + \sin 3v - \sin 3(u+v)], \\ v' &= v + \lambda[\sin 3u - \sin 3(u+v) - 2\sin 3v]. \end{aligned} \tag{12.1}$$

Sum-to-product identities then give

$$\boxed{\zeta' = \zeta - 3\lambda \sin 3\zeta \cos(3v/2)}, \qquad \boxed{v' = v - 2\lambda \sin(3v/2)[\cos 3\zeta + 2\cos(3v/2)]}. \tag{12.2}$$

The sum $\sum_j \Delta_j = 0$ is automatic from pair antisymmetry. Changing lifts by $u \mapsto u + 2\pi a$, $v \mapsto v + 2\pi b$ changes ζ by $2\pi a + \pi b$; the displayed phase rectangles use fixed local lifts, not globally identified real angles.

Where $\lambda > 0$ and $\cos(3v/2) > 0$, the mean moves downward below $\zeta = \pi/3$ and upward above it in the neighboring intervals. The line itself is invariant under synchronization. This statement is about the mean increment; it is neither invariance under $\mathcal{G}$ nor a chiral-area criterion.

Theorem 12.1 (Exact weighted finite-step criterion) *For any positive magnitudes and any real lifted phases, set $w_{jk} = r_j^2 r_k^2$. Then synchronization increases, preserves, or decreases full chiral magnitude according as*

$$\begin{aligned}\mathcal{D} = w_{12}\{\sin^2 u' - \sin^2 u\} + w_{23}\{\sin^2 v' - \sin^2 v\} \\ + w_{13}\{\sin^2(u'+v') - \sin^2(u+v)\}\end{aligned} \tag{12.3}$$

is positive, zero, or negative.

Proof. In cyclic component order,

$$\mathbf{C} = (r_2 r_3 \sin v,\ -r_1 r_3 \sin(u+v),\ r_1 r_2 \sin u).$$

Squaring and subtracting gives exactly $\mathcal{D} = \|\mathbf{C}^+\|^2 - \|\mathbf{C}\|^2$. For bounding any summand use $\sin^2 b - \sin^2 a = \sin(b+a)\sin(b-a)$. Taking square roots preserves the sign of the squared-norm comparison. □

12.2 Explicit closed domains

Write

	u	v	r_1	r_2	r_3
$\mathcal{S}_3 \times \mathcal{A}_3$	$[1.214, 1.223]$	$[.0404, .0447]$	$[.140, .144]$	$[.284, .289]$	$[.362, .369]$
$\mathcal{S}_4 \times \mathcal{A}_4$	$[.587, .614]$	$[.020, .038]$	$[.174, .181]$	$[.328, .335]$	$[.415, .424]$

(12.4)

and $\Lambda = [.16999999, .17000001]$. These are Cartesian product hypotheses, including their endpoints, not correlated samples of a trajectory.

Theorem 12.2 (Uniform gain on the third domain) *For all $(u, v, r, \lambda) \in \mathcal{S}_3 \times \mathcal{A}_3 \times \Lambda$,*

$$\mathcal{D} > .00034. \tag{12.5}$$

On $\mathcal{S}_3 \times \Lambda$ without the magnitude box, a sufficient gain condition is $.1(r_1/r_3)^2 + .08(r_1/r_2)^2 > .002$.

Proof. Here $\zeta \in [1.2342, 1.24535]$ and $3v/2 \in [.0606, .06705]$. Writing $3\zeta = \pi + \alpha_0$ gives $.561 < \alpha_0 < .595$. The alternating polynomial bounds detailed below give

$$.531 < -\sin 3\zeta < .561, \qquad -.847 < \cos 3\zeta < -.828,$$

$$\cos(3v/2) > .9977, \qquad 1.4988 < \frac{\sin(3v/2)}{v} < 1.5.$$

Substitution into (12.2) yields

$$.27018 < \zeta' - \zeta < .28612, \quad 1.5043 < \zeta' < 1.5315, \quad .402 < v'/v < .415,$$

$$.0162 < v' < .0186, \quad 1.495 < u' < 1.524, \quad 1.512 < u' + v' < 1.541 < \pi/2.$$

Let $A_{12} = \sin^2 u' - \sin^2 u$, $A_{13} = \sin^2(u'+v') - \sin^2(u+v)$, and $L_{23} = \sin^2 v - \sin^2 v'$. Bounding the cosine deficits from $\pi/2$ and the small-angle sines gives

$$A_{12} > .3407^2 - .0758^2 = .11033085 > .1, \quad A_{13} > .2984^2 - .0588^2 = .08558512 > .08,$$

$$.0012 < L_{23} < .002.$$

For the last pair, retain $v' < .415v$ and $\sin v > .99966v$: $L_{23} > (.99966^2 - .415^2)(.0404)^2 > .0012$, while $L_{23} < (.0447)^2 < .002$. Thus $\mathcal{D} > .1w_{12} + .08w_{13} - .002w_{23}$, proving the ratio condition. On the full magnitude box the latter expression exceeds

$$.1(.140 \cdot .284)^2 + .08(.140 \cdot .362)^2 - .002(.289 \cdot .369)^2 = .000340818146238 > .00034.$$

All inequalities have strict slack on the closed product. □

Theorem 12.3 (Uniform loss on the fourth domain) *For all* $(u, v, r, \lambda) \in \mathcal{S}_4 \times \mathcal{A}_4 \times \Lambda$,

$$\mathcal{D} < -.00248. \tag{12.6}$$

For every $(u, v, \lambda) \in \mathcal{S}_4 \times \Lambda$ *and* arbitrary *strictly positive magnitudes, the sign remains negative. The fixed absolute margin in* (12.6) *requires the magnitude box.*

Proof. Now $\zeta \in [.597, .633]$ and $3v/2 \in [.030, .057]$. The same polynomial bounds give

$$-.323 < \cos 3\zeta < -.218, \qquad .946 < \sin 3\zeta < .977,$$
$$\cos(3v/2) > .9983755, \qquad \frac{\sin(3v/2)}{v} > 1.49918775.$$

Consequently

$$.0987 < \zeta' < .1514, \quad .091 < v'/v < .147, \quad .00182 < v' < .005586,$$
$$.095 < u' < .152, \qquad .099 < u' + v' < .155.$$

All three positive pair angles contract and remain below $\pi/2$. Quantitatively,

$$-A_{12} > .553^2 - .152^2 > .28, \quad -A_{13} > .5697^2 - .155^2 > .30, \quad -A_{23} > .00039.$$

The last bound uses the correlated inequality $v' < .147v$: $-A_{23} > (.99975^2 - .147^2)(.020)^2 = .000391156425 > .00039$, since $\sin v > .99975v$ throughout this rectangle. Thus $\mathcal{D} < -.28w_{12} - .00039w_{23} - .30w_{13} < 0$ for all positive magnitudes. On $\mathcal{A}_4$ the loss is greater than

$$.28(.174 \cdot .328)^2 + .00039(.328 \cdot .415)^2 + .30(.174 \cdot .415)^2 = .002483531096736 > .00248.$$

□

Written bound method and review status. The theorem proofs use the finite rectangles, not the outcome table. The rational trigonometric bounds are inherited from R8, Section 4, with their sign and endpoint logic reviewed here. For $0 \le t \le .61$, the degree-seven alternating sine polynomial S_7 satisfies $S_7(t) \le \sin t \le S_7(t) + t^9/9!$, with remainder below $3.3\,10^{-8}$; the degree-eight cosine polynomial has error at most $t^{10}/10! < 2\,10^{-9}$. Reduce about π or $\pi/2$ using $3.1415926 < \pi < 3.1415927$ and parity, then apply these polynomials. These estimates justify the outward strict bounds displayed above. The final rational margin expansions are checked in writing in the review notes. Their subsequent whole-domain interval checks are recorded in the completed audit [25]; this document pass runs no evaluator.

12.3 Why the mean line is not the response boundary

Throughout this subsection take $\lambda \in \Lambda = [.16999999, .17000001]$. The weights in (12.3) and the finite phase step both matter. On the invariant line $\zeta = \pi/3$, take $v \in [.02, .04]$ and $r_2 = r_3 = \rho > 0$. Then $.489 < v'/v < .493$ and

$$\mathcal{D} = \rho^2(\cos v' - \cos v)\left[\frac{r_1^2}{2} - \rho^2(\cos v' + \cos v)\right], \qquad 1.9984 < \cos v' + \cos v < 2. \tag{12.7}$$

It is negative for $r_1 = \rho$ and positive for $r_1 = 3\rho$ on the same mean line. These are analytic comparisons of the existing map, not new native trajectories. Likewise, on the slice $v = 0$, $\mathcal{D} = r_1^2(r_2^2 + r_3^2)\sin(\zeta' + \zeta)\sin(\zeta' - \zeta)$. At $\zeta = \pi/2$, the positive mean increment 3λ lowers $\sin^2\zeta$ by overshooting its maximum. A direction of mean movement alone cannot replace the finite-step criterion.

12.4 Separate inclusion of the actual certified inputs

The preceding domain theorems are independent of the four-update trajectory. To apply them to that trajectory, R6–R8 supply additional information. Their comparison pre-sync vectors and the propagated full-box radii imply

	u	v	pre-sync complex-component radius
V_3	$(1.214487, 1.221613)$	$(.040439, .044611)$	$< .00032$
V_4	$(.58711, .61399)$	$(.02075, .03755)$	$< .00151.$

(12.8)

Their magnitudes lie in $\mathcal{A}_3, \mathcal{A}_4$ and each input's coefficient lies in Λ. The intervals are strictly within (12.4). Appendix D exhibits the argument-error allowances. This inclusion inherits the nominal residual chain, separately checked with its original-box application in the subsequent audit [25, 26]. Reuse of R6 in R7 and R8 alone was not that check.

After the third synchronization, $\zeta' > 1.5043 > \pi/3$. The next polynomial stage supplies V_4 with $\zeta < .633 < \pi/3$ before the same synchronizer is applied again. The polynomial update, rather than a change in λ, therefore supplies the different domains. In its first component, the own-state and neighbor real contributions are respectively $(.14999, .15038) + (-.10174, -.10154)$ for V_3, and $(.05675, .05848) + (-.10890, -.10802)$ for V_4. The imaginary components are positive. The first pre-sync phase changes from quadrant I to quadrant II while the other two remain in quadrant II. Arguments of sums are not sums of arguments; the displayed signed sums explain the domain change without making that invalid replacement.

13 Synthesis

The refined geometry provides an open finite surface, positive axial clearance, a precisely located inner area minimum, and distinguishable threefold fan openings and shell notches. At zero it admits a same-hand upper/lower half-turn; away from zero the ring-size obstruction prevents that correspondence within the registered family. Reversing a traversal requires neither reflection nor shape change.

The existing Z readout has its own exact structure. Cone, frozen-curve, and discrete contacts are different questions. A same-state chiral translation permits the exact fan witness, but its four scheduled updates do not describe a continuous passage, and the harmonic return does not restore interior membership. Full-Z correspondence follows explicit residual conditions rather than visual analogy.

The native mechanisms are equally specific: the finite-increment norm remainder matters, polynomial and synchronization effects on chirality can oppose or reinforce each other, and the weighted finite-step criterion has explicit domains of strict gain and loss. The locally uniform example illustrates those mechanisms with the inherited enclosure chain and its completed scoped audit distinguished [25, 26]. No crystal motion law, geometry-to-native feedback, transport equation, SRG premise, or memory premise has been supplied by these results. This manuscript ends at the existing fourth update for review.

A Finite-region tests and the classification margins

A.1 A reusable triangle test

For a triangle $[A, B, C]$, put $b = B - A$, $c = C - A$, $p = P - A$ and $G = (b \cdot b)(c \cdot c) - (b \cdot c)^2 > 0$. A coplanar point belongs to the finite triangle exactly when

$$\lambda_B = \frac{(p \cdot b)(c \cdot c) - (p \cdot c)(b \cdot c)}{G} \geq 0, \quad \lambda_C = \frac{(p \cdot c)(b \cdot b) - (p \cdot b)(b \cdot c)}{G} \geq 0, \quad 1 - \lambda_B - \lambda_C \geq 0.$$

Strict inequalities give its relative interior. This test accompanies every infinite-plane equation used in the text. At the explicit zero cone witness in Section 7, the positive point's weights on $[T_1, T_2, B_3]$ are $((1+\eta_*)/2, (1-3\eta_*)/2, \eta_*)$; the negative point's weights on $[T_0, B_2, B_1]$ are $(\eta_*, (1+\eta_*)/2, (1-3\eta_*)/2)$.

A.2 Section tests for the four-update box

At zero, all section vertices have radius $a\sqrt{1-\eta+\eta^2}$ or the corresponding radius from the other cross-edge; the maximum is $a\sqrt{1-\eta+\eta^2}$. For observation two, the inherited radial lower bound .155 already exceeds a by more than .0085, excluding the whole band.

At observation three, the inherited Cartesian box is

$$X_3 \in (.0005, .0014), \quad Y_3 \in (-.1400, -.1387), \quad w_3 \in (-.0939, -.0926).$$

It gives $\eta_3 \in (.723, .727)$ and an outer section radius less than .1313. The actual point radius exceeds .1387, leaving more than .0074 of radial separation.

At observation four,

$$X_4 \in (-.0675, -.0636), \quad Y_4 \in (.1073, .1111), \quad w_4 \in (.1333, .1370), \qquad \eta_4 \in (.169, .179).$$

A global outer-radius test alone is insufficient. Its ray meets the finite section edge D_3S_4 of $[T_3, T_4, B_5]$, since

$$D_3 = a(-k_0\eta, 1-1.5\eta), \qquad S_4 = a(-k_0, .5-\eta), \qquad k_0 = \sqrt{3}/2,$$

$$\det(D_3, (X_4, Y_4))/a > .7315(.0636) - .8661(.179)(.1111) > .029,$$

$$\det((X_4, Y_4), S_4)/a > .8660(.1073) - .0675(.331) > .0705.$$

These signs place the ray between the endpoints. The outward unit normal $n = (-1/2, k_0)$ then gives

$$n \cdot (X_4, Y_4) > .1247218, \qquad n \cdot D_3 = n \cdot S_4 = ak_0(1-\eta) < .1055.$$

The residual exceeds .0192. This proves exterior status in the correct finite angular sector, with radial separation greater than .0192. It is not a claimed minimum distance to the entire band.

For the first native-only endpoint R4 gives

$$X_N \in (.14847, .14900), \quad Y_N \in (-.04516, -.04479), \quad w_N \in (.19192, .19245),$$

so $R_N > a + .00197$. The scheduled first endpoint satisfies $X_1 \in (.10550, .10603)$, $Y_1 \in (.02344, .023895)$ and the height in Table 2; its radius is $R_1 < .109$. At each input's own perturbed height put $\eta_1 = (1 - w_1/h)/2$. For the actual zero-member section, the two triangle types give

$$\mathcal{R}(w, \varphi) \geq L(\eta) = \min\left\{\frac{a\sqrt{3}(1-\eta)}{2}, \, a\left(1 - \frac{3}{2}\eta\right)\right\}. \tag{A.1}$$

On a top-two-vertex triangle the projection on its unit horizontal plane normal is $a\sqrt{3}(1-\eta)/2$; on a bottom-two-vertex triangle the projection on e_i is at least $a(1-3\eta/2)$. These projections bound every section segment, including the inward vertices. Since the established star-shaped section contains the axis, its filled interior contains the open disk of radius $L(\eta)$ whenever both bounds are positive. The inherited height and radical bounds give $0 < \eta_1 < .097$ and

$$\frac{a\sqrt{3}(1-\eta_1)}{2} > .1464(.8660)(.903) > .1144, \qquad a(1-3\eta_1/2) > .1464(.8545) > .1144.$$

Thus $L(\eta_1) > .1144 > R_1$, with an interior clearance greater than .0054, at every input-dependent height and azimuth. This restores R4 Section 5.4, equations (32)–(34); its numerical substitutions remain inherited from R4 and subsequently checked [19, 25]. The disk is a consequence of the actual triangles, not a replacement geometry. Both endpoints are inside the host because their radii are less than r, and both satisfy $0 < w_1 < w_N < w_0$.

For all scheduled $n = 1, \dots, 4$, $R_n < r$ and $|w_n| < h$ prove host central-prism membership. Fan and notch points have $|w| \geq h$, so these observations cannot lie on those finite sets. Their axis and center distances are bounded from the Cartesian boxes by $\sqrt{X_n^2 + Y_n^2}$ and $\sqrt{X_n^2 + Y_n^2 + w_n^2}$ using rational squaring. The distance intervals in the main table are intentionally wider than the tightest possible ones.

B The inherited four-update enclosure chain

B.1 Status and exact starting point

This appendix records the load-bearing numerical chain of R3–R6. The subsequent standalone audit checked its initial inclusion, comparison residuals, nonzero-domain coverage, sensitivity estimates, and propagation recurrences; GPT's unchanged-checker replay matched the saved records [25, 26]. The original bounds and source attributions are retained. No change has been made to the original input uncertainty.

The exact initial state is (8.2). A cancellation-free form is

$$x = \left(-\frac{Xc_2^\star}{b_\perp}, \frac{Xc_1^\star}{b_\perp}, 0 \right), \quad y = \left(-\frac{c_3^\star c_1^\star}{Xb_\perp}, -\frac{c_3^\star c_2^\star}{Xb_\perp}, \frac{b_\perp}{X} \right). \tag{B.1}$$

R3 bounds

$$1.4142135 < \sqrt{2} < 1.4142136, \quad 1.7320508 < \sqrt{3} < 1.7320509,$$
$$.7438391 < c < .7438393, \quad .6683 < s_\ell < .6684, \quad .9439 < E < .9440,$$
$$c_1^\star \in (-.6049883, -.6049877),\ c_2^\star \in (-.3388968, -.3388966),\ c_3^\star \in (-.0800636, -.0800629),$$
$$b_\perp \in (.69344, .69345), \qquad X \in (2.99090, 2.99092).$$

Roots are bounded by squaring rational endpoints. Cosine at .732 uses alternating terms through degrees ten and eight, sine through seven and nine, and $\exp(-.0577)$ through five and four. These are exact rational arguments. R6 refines (B.1) to

	1	2	3
x	$(1.461705, 1.461710)$	$(-2.609399, -2.609390)$	0
y	$(-.0233545, -.0233541)$	$(-.0130826, -.0130822)$	$(.2318496, .2318503)$.

B.2 Comparison centers and pre-sync residuals

The following terminating decimals are rational *comparison vectors*, not replacement initial data. A pair (a, b) in this table denotes $a + \mathrm{i}b$, not an interval.

For example, at the exact initial state the nominal real matrix is

$$\mathbf{B}_* = \begin{pmatrix} .8 - .1I_1 & .2 & .2 \\ .2 & .9 - .1I_2 & .2 \\ .2 & .2 & 1 - .1I_3 \end{pmatrix}.$$

n	$\widehat{\Omega}_{n,1}$	$\widehat{\Omega}_{n,2}$	$\widehat{\Omega}_{n,3}$
0	$(1.461707, -.0233543)$	$(-2.609395, -.0130824)$	$(0, .2318499)$
1	$(.322706, .095167)$	$(-.259081, .111445)$	$(-.244759, .206520)$
2	$(.188739, .085078)$	$(-.223602, .148232)$	$(-.284600, .179065)$
3	$(.072196, .121963)$	$(-.237006, .161219)$	$(-.305291, .199972)$
4	$(-.102612, .144250)$	$(-.223813, .244521)$	$(-.284466, .308622)$

Table 5: R6 rational comparison centers, with the exact initial state within .000006 in maximum complex-component norm.

n	$\widehat{V}_{n,1}$	$\widehat{V}_{n,2}$	$\widehat{V}_{n,3}$
1	$(.3351004, .0300612)$	$(-.2793475, .0388329)$	$(-.2295376, .2233163)$
2	$(.153744, .138649)$	$(-.215523, .159751)$	$(-.229524, .245724)$
3	$(.048542, .133157)$	$(-.218805, .185171)$	$(-.288355, .223702)$
4	$(-.050848, .169564)$	$(-.257977, .208159)$	$(-.334187, .253945)$

Table 6: R6 pre-sync comparison centers. Each real and imaginary difference from $\mathcal{G}_{\mathcal{D}_*}(\widehat{\Omega}_{n-1})$ is less than .000002; the complex-component difference is less than .000003. These are inherited polynomial residual bounds.

R3's wider direct certificate gives

$$V_{1,1} \in (.33497, .33523) + \mathrm{i}(.03005, .03008),$$
$$V_{1,2} \in (-.27953, -.27916) + \mathrm{i}(.03882, .03885),$$
$$V_{1,3} \in (-.22958, -.22950) + \mathrm{i}(.22330, .22334).$$

It can use $\sin 3\varphi = 3\sin\varphi - 4\sin^3\varphi$ on normalized pair products instead of phase inversion. The refined R6 route below uses small arctangent arguments to sharpen subsequent stages.

B.3 Argument and synchronization residual allowances

At the exact polynomial image of each comparison center, R6 encloses the three small arguments t_j as follows: The phase triples are respectively

n	t_1	t_2	t_3
1	$(.089706, .089711)$	$(.139012, .139016)$	$(.013736, .013741)$
2	$(.051620, .051628)$	$(.148613, .148624)$	$(.034083, .034092)$
3	$(.36453, .36457)$	$(.08325, .08327)$	$(.12625, .12628)$
4	$(.29986, .29989)$	$(.10686, .10689)$	$(.13642, .13646)$

Table 7: Inherited argument quotients at the exact polynomial images, not at the rounded $\widehat{V}_n$.

1	$\arctan t_1$	$\pi - \arctan t_2$	$3\pi/4 + \arctan t_3$
2	$\pi/4 - \arctan t_1$	$3\pi/4 + \arctan t_2$	$3\pi/4 - \arctan t_3$
3	$\pi/2 - \arctan t_1$	$3\pi/4 + \arctan t_2$	$3\pi/4 + \arctan t_3$
4	$\pi/2 + \arctan t_1$	$3\pi/4 + \arctan t_2$	$3\pi/4 + \arctan t_3$.

The alternating arctangent polynomial through degree thirteen has error less than $t^{15}/15 < 2\,10^{-8}$ for $t \le .365$. Reduce the triple angles about multiples of $\pi/4$; residual arguments are less than 1.5. Sine through degree thirteen and cosine through twelve have errors less than $4\,10^{-9}$ there. The same bound suffices for the smaller final increments. The resulting allowances are

n	Δ_1	Δ_2	Δ_3	$\rho_n > \left\|\mathcal{F}_{\mathcal{D}_*}(\widehat{\Omega}_{n-1}) - \widehat{\Omega}_n\right\|_\infty$
1	$(.19729, .19733)$	$(-.26812, -.26808)$	$(.07077, .07082)$	.00002
2	$(-.31035, -.31030)$	$(.05243, .05248)$	$(.25784, .25791)$	.00004
3	$(-.18495, -.18485)$	$(.10496, .10504)$	$(.07987, .07995)$	.00005
4	$(.32690, .32703)$	$(-.15071, -.15060)$	$(-.17636, -.17625)$	.00005

Table 8: Inherited simultaneous-increment and full-update residual bounds. The displayed norm is maximum complex-component norm.

B.4 Domains and Lipschitz propagation

For an own-channel polynomial component, the real two-dimensional radial and tangential derivative eigenvalues are $1 - 2g + \epsilon(k_j - 3I_j)$ and $1 - 2g + \epsilon(k_j - I_j)$. Together with the off-diagonal g terms, these give a component maximum-norm Lipschitz constant 1.6 initially and 1.41 in the later domains.

For synchronization, let $a_{jm} = \cos 3(\phi_m - \phi_j)$. The row bound used in R6 is

$$\max\left\{1, \left|1 - 3\lambda \sum_{m\neq j} a_{jm}\right|\right\} + 3|\lambda| \sum_{m\neq j} \frac{|V_j|}{|V_m|}|a_{jm}|. \tag{B.2}$$

It follows by differentiating radial and angular coordinates on a convex nonzero neighborhood, with $|d\phi_m| \le |dV_m|/|V_m|$. The relevant domains are

n	(r_1, r_2, r_3) intervals	a_{12}	a_{13}, a_{23}	L_n
1	all $(.28, .34)$	$[-1, 1]$	$[-1, 1], [-1, 1]$	4
2	$(.206, .209), (.267, .270), (.334, .338)$	$(.45, .65)$	$(-.10, .10), (.80, .90)$	2
3	$(.140, .144), (.284, .289), (.362, .369)$	$(-.91, -.82)$	$(-.85, -.73), (.97, 1]$	3.2
4	$(.174, .181), (.328, .335), (.415, .424)$	$(-.29, -.16)$	$(-.39, -.23), (.98, 1]$	2.6

Table 9: R6 nonzero domains and synchronizer constants. Their coverage must be checked together with the recurrences, not assumed from later outcomes.

Let e_n bound nominal distance from the comparison center. The recurrence $e_n \le L_n L_G e_{n-1} + \rho_n$ admits

$$(e_0, e_1, e_2, e_3, e_4) = (.000006, .000060, .000220, .001050, .003900).$$

The right sides before outward rounding are .0000584, .0002092, .00104264, .0038993.

Now let $\delta_0 = 10^{-8}$ denote the *original* input-coordinate radius. The initial component perturbation is at most $\sqrt{2}\delta_0$. R6's first-stage coefficient derivative sum is less than $21.27\delta_0$, and the state term less than $2.4\delta_0$. It takes $D_1 = .00000025$ and $Q_1 = .0000011$, including the $.68\delta_0$ synchronizer-coefficient contribution. At subsequent stages, the combined parameter derivative bound is less than $1.9 + 1.8 + .046 < 4$. Hence

$$D_n \le 1.41 Q_{n-1} + 4\delta_0, \qquad Q_n \le L_n D_n + 2M_n\delta_0, \tag{B.3}$$

where M_n bounds pre-sync component magnitude. Valid outward choices are

n	D_n	Q_n
2	.0000016	.0000033
3	.0000047	.0000151
4	.0000214	.0000557.

The total radii about the rounded pre-sync centers, including the polynomial residual .000003, are

$$1.41(.000060 + .0000011) + .00000004 + .000003 = .000089191 < .00009,$$
$$1.41(.000220 + .0000033) + .00000004 + .000003 = .000317893 < .00032,$$
$$1.41(.001050 + .0000151) + .00000004 + .000003 = .001504831 < .00151.$$

The last value corrects R6's printed .001504871, which was a slightly larger conservative number; the domain containment is unchanged.

B.5 Readout bounds and the earlier coarse proof route

Norm preservation by synchronization makes the macro perturbation depend on the smaller pre-sync radius. For the present weights, each observation-component error is bounded by $1.236D_n + M_nQ_n$; it is less than .000012 for $n = 3$ and .000051 for $n = 4$. Representative nominal scalar bounds are

	$n = 3$	$n = 4$
κ_n	$(.4845, .4860)$	$(.5603, .5664)$
K_n (uniform)	$(.20159, .20221)$	$(.22187, .22353)$
positive $K_nE^n\lvert f_n\rvert$	$(.1132, .1138)$	$(.1309, .1322)$.

Uniform $\kappa_3 \in (.48449, .48601)$ and $\kappa_4 \in (.56025, .56645)$ incorporate the box error. Also $K_2 \in (.19919, .20011)$, $E^3 \in (.8409, .8413)$, $E^4 \in (.7937, .7942)$, and $\sqrt{3}/2 \in (.8660254, .86602545)$. The comparison chiral vectors are

$$\widehat{C}_3 = (.001824145897, -.051671384745, .040545329702),$$
$$\widehat{C}_4 = (.000484295100, -.009365899836, .007194236398).$$

Nominal enclosures are $C_3 \in (.0011, .0026) \times (-.0523, -.0511) \times (.0400, .0411)$ and $C_4 \in (-.0025, .0035) \times (-.0118, -.0070) \times (.0051, .0093)$. Combining them with the amplitude and phase factors produces the Cartesian boxes in Appendix A.

R4 and R5 used broader bounds for the first two observations; the main table retains those conservative early results rather than silently replacing prior reports. In R4 the first pre-sync error is $26\delta_0 + 52\delta_0 = .00000078$; the post-sync error is $21(.00000078) + \delta_0 = .00001639$. The observation error is at most $1.236(.00000078) + .34(.00001639) = .00000653668 < 10^{-5}$; the initial error is at most $9\delta_0$. Adding this allowance to the nominal native-only and scheduled endpoints gives the first-step intervals in Appendix A. R5 uses $D_2 = .00004$, $Q_2 = .00096001$, and observation error $.0003758434 < .0004$. Its nominal second point lies in $(-.0661, -.0641) \times (-.1446, -.1427) \times (-.1098, -.1079)$; the larger uniform box is $(-.0665, -.0637) \times (-.145, -.1423) \times (-.1102, -.1075)$. These proof widths are not new perturbations of the original inputs.

C Mechanism bounds and unresolved direction projections

R7 evaluates the exact budget (11.1) using the comparison states and total radii $t_n = e_n + Q_n = (.0000611, .0002233, .0010651)$ for $n = 1, 2, 3$. The rational comparison intensities are

n	$\hat{I}_1$	$\hat{I}_2$	$\hat{I}_3$
1	.113195920325	.079542952586	.102557478481
2	.042860676205	.071970580228	.113061434225
3	.020087235785	.082163409997	.133191395465.

The elementary bound $|I_j - \hat{I}_j| \le 2|\widehat{\Omega}_j|t_n + t_n^2$ (or $2M_jt_n$ with M_j bounding the whole segment) propagates into onsite channel errors

$$(.000017, .000026, .000034), \quad (.000038, .000073, .000122), \quad (.000123, .000370, .000630),$$

including parameter effects less than $2\,10^{-8}$. For a pair difference, the square error is at most $4|\widehat{\Omega}_j - \widehat{\Omega}_k|t_n + 4t_n^2$. The comparison pair sums are .682401478796, .411568322980, .251893385824. The two growth stages have individual pair-square enclosures

	12	13	23
Ω_2	(.17363, .17440)	(.23245, .23332)	(.00460, .00474)
Ω_3	(.09581, .09849)	(.14693, .15024)	(.00582, .00651).

Their onsite channel contributions are (.01673, .01683), (.04206, .04224), (.08775, .08804) and (.00782, .00809), (.04756, .04834), (.10236, .10366), respectively.

For the finite-increment remainder use $\widehat{D}_n = \widehat{V}_{n+1} - \widehat{\Omega}_n$. The component error is $b_n \le 2.41t_n + 4\delta_0 + .000003$, with outward values .000150291, .000541193, .002569931. The comparison squared norms are .036438669609, .025360851774, .023796891123 and the sums of component moduli are less than .284, .232, .246. Thus the squared-norm errors are at most $2b_n \sum_j |\widehat{D}_{n,j}| + 3b_n^2$, bounded by .000086, .000252, .001285. These allowances yield Table 3. The mixed term in (11.2) lies in (−.07209, −.07161), (−.04950, −.04836), and (−.03624, −.03203).

For chiral stages, bilinearity and (11.4) give the component enclosures inherited from R7:

	C_1	C_2	C_3
Ω_2	(.00201, .00229)	(−.05814, −.05788)	(.04689, .04712)
V_3	(.00423, .00466)	(−.04942, −.04909)	(.03798, .03827)
Ω_3	(.00112, .00253)	(−.05222, −.05112)	(.04008, .04101)
V_4	(.00290, .00520)	(−.04467, −.04283)	(.03237, .03394)
Ω_4	(.00024, .00080)	(−.01177, −.00697)	(.00515, .00924).

The broad Cartesian boxes alone do not prove every direction comparison. The ratio

$$\frac{C_1}{C_3} = \frac{r_3}{r_1}\frac{\sin v}{\sin u}$$

at positive u, v retains correlations during synchronization. R7 gives C_1/C_3 in (.042, .049) for Ω_2, (.110, .123) for V_3, (.027, .064) for Ω_3, and (.085, .161) for V_4. The separated adjacent intervals prove changes at both polynomial stages and at the third synchronization. At the fourth synchronization,

$$\frac{(C_{4,1}/C_{4,3})}{(C_{V,4,1}/C_{V,4,3})} = \frac{\sin v'}{\sin v}\frac{\sin u}{\sin u'} < \frac{.147}{.99975}\frac{.614}{.9962(.097)} < .935 < 1.$$

This uses the correlated R7 refinement $.097 < u' < .149$. It follows from the comparison bound $\hat{u}' \in (.12266, .12320)$ and $|u' - \hat{u}'| < 1.466(.0133) + .702(.0083) + 4\delta_0 < .025325$. The derivative constants apply on the full fourth phase rectangle.

For the fourth vertical fraction, the refined magnitude bounds $r_1 \in (.17551, .17854)$, $r_2 \in (.32997, .33300)$, $r_3 \in (.41821, .42124)$ give $-C_{4,2}/C_{4,3} \in (1.270, 1.351)$ and $0 < C_{4,1}/C_{4,3} < .139$ using $\sin(u' + v')/\sin u' = \cos v' + \cot u' \sin v'$. Then $C_3/\|\mathbf{C}\| = [1 + (C_1/C_3)^2 + (C_2/C_3)^2]^{-1/2}$ gives the fraction in Table 4. These refinements retain their R7 Sections 5.3–6.2 dependency, subsequently checked in the audit [25], and must not be replaced by the wider independent-product bounds of Section 12.2. The third synchronization's fraction change and all fourth-stage fraction changes remain unresolved, as does the net direction change from three to four. No direction claim here rests on treating interval overlap as proof of equality or heterogeneous behavior.

D Actual-image inclusion in the phase domains

R8 Section 6 uses pre-sync radii .00032 and .00151 from Appendix B. The relevant comparison phases satisfy

$$\hat{u}_3 \in (1.2179, 1.2182), \quad \hat{v}_3 \in (.04245, .04260), \qquad \hat{u}_4 \in (.6004, .6007), \quad \hat{v}_4 \in (.0290, .0293).$$

On each nonzero complex ball, integrate $|d \operatorname{Arg} z| \le |dz|/|z|$ along a segment. Dividing the radii by the positive domain magnitude minima gives per-channel phase errors less than

$$(.002286, .001127, .000884) \quad \text{and} \quad (.00868, .00461, .00364),$$

respectively. Adding the two appropriate errors to each relative phase gives exactly the intervals in (12.8). Squaring the Cartesian component boxes gives the amplitude boxes in (12.4); all exclude zero. This establishes inclusion for the unchanged closed twelve-coordinate input product using the inherited nominal residual chain, now separately checked with the inclusion [25, 26].

For the phase theorems themselves, the absolute margin multiplications were checked separately from that chain. The gain terms are $.00015808576 + .000205476992 - .000022744605762 = .000340818146238$. The loss terms are $.00091201969152 + .000007226175216 + .00156428523 = .002483531096736$. These exact terminating-decimal sums were originally checked in writing and subsequently verified by the standalone audit [25]. The wider theorem rectangles and the actual-image intervals continue to serve different logical purposes.

E Exceptional symmetry cases and finite contact transport

This appendix completes the grid classification of Proposition 10.1 from R9 Sections 5–7. Assume $A > 0$, $\alpha > 0$, and $\beta > 0$. For Q with shell action $-P$, the odd-sector conditions are $C_1 = C_3$, $C_2 = 0$. At even sectors the sum of the first and third components of the residual requires $\cos\theta = -1$, so only sector six can work. Its remaining condition is $C_2 = 0$, $C_1 - C_3 = -2\alpha Ac/\beta$. Sector zero cannot work for a nonzero state, and no nonzero all-grid identity results.

For S with $-P\mathcal{K}$, odd sectors require $C_1 = C_3 = 0$. At even sectors the second component forces $\sin\theta = 0$. At sector zero the conditions are $C_1 = 0$, $C_3 = -2\alpha Ac/\beta$; at sector six they are $C_3 = 0$, $C_1 = -2\alpha Ac/\beta$. For Q with the alternative conjugation action, (10.3) has an uncancellable third macro component at every even sector, so only the odd-sector condition $C_2 = 0$ remains.

For R, solving $P_c\mathbf{C} = R\mathbf{C}$ gives $C_2 = C_3$ and $C_1 = -(2 + \sqrt{3})C_2$. Imposing the corresponding R^2 equation as well leaves only zero. The common-phase restrictions described in Section 10 apply to native-map covariance, not to these algebraic residual formulas.

If $\mathbf{z}$ has barycentrics (b_1, b_2, b_3) on a finite face, then $G\mathbf{z}$ has the same barycentrics on the image face under an orthogonal G. For notches, the index actions following (5.2) preserve the bounds in (4.1), possibly reversing the transverse parameter. These facts prove finite contact transport when the relevant residual is zero. They make no contact-existence assertion. Equal-time state transformations, clock relabels, and forward native evolution remain three separate operations.

F Source crosswalk and review dependencies

Source notation	Here	Reason
Crystal t in R0–R2, R9	τ	Distinct from observer time t_c.
Planar family matrix A_t	A_τ	Distinct from scalar amplitude A.
Section fraction in R1–R2	η	Not synchronizer coefficient λ.
Area A in R2	$\mathcal{A}$	Distinct from the macro amplitude.
State norm in all reports	κ	Native norm, not plotted radius.

Source notation	Here	Reason
Plot v in R2–R7	$\mathbf{z}$	Keeps relative phase v unambiguous.
Phase mean ζ in R8	ζ	Unchanged; not observation or height.
Chiral C_3 and stage n	$C_{n,3}$ when needed	Component and update index separated.
R3 witness c_*, t, p, q	$c^\star, b_\perp, p_0, q_0$	No collision with clock or family parameter.
Height reflection in R9	H	Height itself is w; tip height is H_0.
R6 e_n, D_n, Q_n	same, explicitly typed	Proof errors and input-box errors distinguished.

The geometric definitions, section determinant proof, cone monotonicity, state/readout dependence, phase reduction, weighted criterion, and symmetry residual signs have been reviewed as written algebra for this draft. The finite phase-domain proofs reproduce the original inequalities and remainder method independently of the trajectory table. The final strict margin sums and corrected R6 coverage value have been checked in writing.

The later standalone audit checked the required inherited numerical dependencies, including the entire original input box through four updates, each input's fixed configuration, finite-region tests, mechanism bounds, and the full closed phase/amplitude products [25]. All six numerical proof blocks passed: 768 scalar bound records, with 58 arithmetic self-checks and eight exact supporting-identity checks counted separately. Of the 37 claim-map rows, 14 have quantitative checks within scope, six are checked only as supporting definitions or identities, and 17 lie outside the numerical audit. These counts do not represent independent theorems, and overlapping per-claim records are not added.

GPT inspected the checker and proof structure and replayed the same unchanged checker in an isolated environment. All 768 scalar records matched exactly, including endpoints, margins, and statuses [26]. This is replay of the standalone audit, not a third independently written evaluator or independent human peer review. Outward interval evidence supports whole-domain claims; the high-precision nominal calculation is corroboration only. The certificate remains conditional on the inspectable interval implementation, integer and rational arithmetic, correct source transcription, and the stated analytic remainder bounds. It is not proof-assistant verification, production binary64 certification, or experimental validation. The existing unresolved direction/projection comparisons remain unresolved. This document-only finalization adds no scientific execution, tighter bound, or new conclusion.

Two editorial points are recorded without changing earlier reports. First, the R6 coverage arithmetic is .001504831, as displayed above. Second, the notch formula contains the addition sign in (4.1). The manifest-identified R9 equation (11) omits the plus sign before its transverse term. This manuscript restores it using R2 equations (7)–(8). The v0.1 account that the local R9 copy already contained the sign was incorrect; a fresh byte and hash comparison confirms the omission. R9 remains unchanged, and no geometry definition has been changed.

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